TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Temperature-Triggered Extracellular Glutathione Production: Eliminating Cell Lysis to Reduce Manufacturing Costs**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( ) *(Please choose at least one of the areas above)*			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio-Medical		Chemicals	
	Medium Category	Small Category	Medium Category	Small Category
	Industrial biotechnology	Bio-chemical materials	Fine chemicals	Pharmaceutical intermediates / active ingredients
Concept of Proposed Technology	Glutathione (GSH) production currently requires expensive cell lysis and purification, accounting for 60-80% of manufacturing costs in the $250M+ global market. Current industrial processes must lyse cells to extract intracellular GSH, necessitating complex downstream processing that dramatically increases costs and generates substantial waste. We have discovered that the budding yeast, *Saccharomyces cerevisiae*, secretes GSH extracellularly when cultured above 36.5°C—without cell death or lysis—enabling direct harvesting from culture			

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	supernatant (Laman Trip & Youk, *Nature Microbiology* 2020). Our preliminary data show ~10 μM GSH secreted from 10^6 cells/mL at 39°C within 48 hours, with cells remaining viable throughout. This eliminates intentional cell lysis, reducing production costs by an estimated ~50% or more, with technical and economic analyses suggesting a path from current lysis-based costs (~$150-200/kg) toward substantially lower-cost production. This project will: (1) systematically optimize secretion rates across temperature ranges and growth conditions, (2) engineer strains for 5-10x enhanced GSH production and secretion, (3) develop scalable bioprocessing protocols using standard fermentation equipment, and (4) demonstrate pilot-scale production with Korean SME partners. This technology positions Korean biotech companies competitively in the rapidly growing global GSH market (7-9% annual growth), particularly leveraging Korea's strengths in fermentation technology and cosmetics manufacturing (K-beauty applications).
Rationale for Joint R&D Project	While GSH demand continues to grow globally, current production methods rely heavily on intracellular accumulation followed by cell lysis, increasing cost and complexity. This joint R&D project aims to translate a **fundamentally different production paradigm**—non-lytic secretion—into an industrially viable process. Collaboration with Korean SMEs is essential to optimize fermentation conditions, scale-up strategies, and downstream purification tailored to commercial manufacturing. SMEs with expertise in fermentation engineering, purification materials, or bioprocess automation would benefit from access to a novel production mechanism with strong cost-reduction potential.
Keywords	1. Glutathione (GSH) 2. Yeast bioprocessing 3. Non-lytic secretion 4. High-temperature fermentation 5. Antioxidant production

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3 - 4)	End of the Project	Level (5 - 6)
R&D Objectives	**Current Technology Status:** • Temperature-triggered GSH secretion demonstrated in wild-type *S. cerevisiae* (Nature Microbiology 2020) • Proof-of-concept: ~10 μM GSH from 10^6 cells/mL at 39°C within 48 hours • Cells remain viable during secretion (no lysis required) • Secretion threshold identified: >36.5°C **Technology Gaps to Address:** • Systematic optimization of secretion conditions not yet performed • Secretion rates insufficient for commercial-scale production • Strain engineering needed for enhanced GSH production and secretion • Scalable bioprocess protocols not yet developed • Pilot-scale demonstration with industrial partner needed **R&D Objectives:** 1. Systematically optimize GSH secretion rates across temperature (36-42°C), cell density (10^5-10^8 cells/mL), growth phase, and medium composition 2. Engineer *S. cerevisiae* strains with 5-10x enhanced GSH production through overexpression of biosynthesis genes (GSH1, GSH2) and optimized transporter expression. Initial optimization will prioritize process conditions and native regulation, with genetic modifications introduced only where necessary for productivity gains. 3. Develop scalable two-stage bioprocess: growth phase (30°C) followed by secretion phase (39°C) 4. Demonstrate pilot-scale production (50-100L) with Korean SME partner, achieving ≥50% cost reduction vs. conventional lysis-based methods			
R&D Target Specifications	**Phase 1 (Months 1-6): Process Optimization** • Achieve 3-5x improvement in GSH secretion through systematic optimization • Identify optimal conditions: temperature (±1°C precision), cell density, timing • Demonstrate reproducibility at 1-5L scale • Target: 30-50 μM GSH from 10^6 cells/mL **Phase 2 (Months 7-12): Strain Engineering** • Develop engineered S. cerevisiae strains with enhanced GSH biosynthesis • Target: 5-10x improvement over wild-type (50-100 μM from 10^6 cells/mL) • Maintain cell viability >95% during secretion phase • Characterize genetic stability over multiple generations **Phase 3 (Months 13-18): Bioprocess Development** • Scale from 1L to 10-50L with <20% productivity loss • Develop two-stage fed-batch protocol (growth + secretion phases) • Optimize cell retention and continuous GSH harvesting systems • Demonstrate integration with standard fermentation equipment **Phase 4 (Months 19-24): Pilot Demonstration** • 50-100L pilot fermentation at Korean SME partner facility • Achieve production cost of $60-90/kg (vs. $150-200/kg conventional) • Demonstrate ≥50% cost reduction with rigorous technoeconomic analysis			

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	• Product purity meets industry standards for target applications (cosmetics, nutraceuticals) • Secure pathway to commercial scale-up (500L-1000L)
R&D Plan	**Aim 1: Systematic Process Optimization (Months 1-6)** Characterize GSH secretion as function of: • Temperature range: 36-42°C (1°C increments) to identify optimal conditions • Cell density: 10^5 to 10^8 cells/mL to maximize volumetric productivity • Growth phase: exponential vs. stationary phase timing for secretion induction • Medium composition: glucose, amino acids (especially cysteine, glutamate), trace elements • Time-course dynamics: up to 7 days to assess sustained secretion Deliverable: Optimized protocol achieving 3-5x improvement; reproducibility demonstrated at 1-5L scale **Aim 2: Strain Engineering for Enhanced Production (Months 7-12)** Engineer *S. cerevisiae* incorporating: • GSH biosynthesis gene overexpression (GSH1: γ-glutamylcysteine synthetase; GSH2: glutathione synthetase) • Enhanced precursor supply (CYS3, CYS4 for cysteine; GLT1, GLN1 for glutamate pathways) • Putative GSH transporter optimization based on mechanistic studies • Reduced intracellular degradation (deletion of γ-glutamyl transpeptidase if necessary) Screen multiple strain constructs; select top 2-3 candidates Deliverable: Engineered strains producing 50-100 μM GSH (5-10x improvement); genetic stability confirmed **Aim 3: Scalable Bioprocess Development (Months 13-18)** Develop production protocols: • Two-stage operation: biomass accumulation (30°C, optimal growth) → GSH secretion (39°C, triggered production) • Fed-batch optimization: nutrient feeding strategies to sustain secretion without cell division • Cell retention systems: enable continuous GSH harvesting while retaining viable cells • Downstream processing: supernatant clarification (filtration), concentration, preliminary purification • Process control: temperature shifts, dissolved oxygen, pH optimization Stepwise scaling: 1L → 5L → 10L → 50L Deliverable: Scalable protocol maintaining ≥80% productivity across scales; integration with standard equipment demonstrated **Aim 4: Pilot Demonstration with Korean SME (Months 19-24)** Partner selection criteria: Existing fermentation capacity (10-100L), yeast experience, downstream processing capability, market access (cosmetics or nutraceuticals) • On-site 50-100L pilot runs at partner facility (minimum 3 batches) • Process transfer and operator training • GSH recovery, purification, quality testing (purity, identity, activity) • Detailed cost analysis: raw materials, utilities, labor, equipment, waste disposal • Compare with conventional lysis-based production (literature + partner's existing processes) • Product qualification: test samples with end customers (cosmetics formulations, nutraceutical ingredients) Deliverable: Demonstrated ≥50% cost reduction; product meeting industry specifications; partnership agreement for commercialization; pathway to commercial scale identified

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**Expected Roles of My Research Team at UIUC:** **Technical Development:** • Systematic optimization of GSH secretion conditions (temperature, density, medium composition) • Strain engineering: design, construction, and characterization of enhanced GSH-producing strains • Bioprocess development: fed-batch protocols, cell retention systems, downstream processing optimization • Analytical methods: GSH quantification, purity analysis, cell viability assays • Scale-up support: troubleshooting during pilot demonstrations, process optimization **Intellectual Property:** • File and prosecute patent applications (process patents and engineered strain IP) • Manage IP portfolio and licensing negotiations • Provide freedom-to-operate analysis **Knowledge Transfer:** • Technical training for SME partner personnel (fermentation protocols, analytical methods) • Document standard operating procedures (SOPs) and process specifications • Ongoing technical consultation during scale-up and commercialization phases • Joint publications highlighting technology advantages (with SME co-authorship) **Project Management:** • Coordinate research activities across Aims 1-4 • Regular progress reporting to TIPA and SME partners • Facilitate communication between UIUC team and Korean SME • Organize site visits and technology demonstration events
Expected R&R of Korean SME Partner	**Infrastructure and Resources:** • Provide pilot-scale fermentation facility (50-100L minimum capacity) • Supply necessary raw materials (media components, gases, etc.) • Offer downstream processing equipment (centrifuges, filters, concentration systems) • Provide analytical instrumentation for product characterization **Technical Expertise:** • Share industrial fermentation expertise and best practices • Contribute knowledge of regulatory requirements for target markets (cosmetics, nutraceuticals, pharma) • Provide insights on customer requirements and product specifications • Assist with process economics analysis (realistic cost projections) **Pilot Demonstration:** • Host on-site pilot runs at their facility (Months 19-24) • Assign experienced personnel to work alongside UIUC team during technology transfer • Conduct product qualification testing with potential customers • Provide feedback on practical implementation challenges and solutions **Commercialization:** • Develop market entry strategy for GSH products • Establish customer relationships and sales channels • Scale technology to commercial production (500L-1000L+) post-project

	• Provide ongoing feedback for continuous improvement • Co-invest in commercial-scale equipment and facility modifications **Ideal Partner Profile:** • Existing yeast fermentation capacity and expertise • Established downstream processing capabilities • Customer relationships in cosmetics, nutraceuticals, or pharma ingredients • Willingness to co-invest in scale-up activities • Located in Korea with access to Asian markets

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technical Outcomes:** • Optimized GSH production process achieving 50-100 μM secreted GSH (5-10x improvement over baseline) • Engineered S. cerevisiae strains with enhanced and stable GSH production • Scalable bioprocess protocols demonstrated from bench (1L) to pilot (50-100L) scale • ≥50% reduction in production costs: from $150-200/kg to $60-90/kg • Product quality meeting industry specifications for cosmetics and nutraceutical applications • Patent portfolio covering process technology and engineered strains **Economic Impact:** • Enable Korean SME entry into $250M+ global GSH market with competitive cost structure • Create new revenue streams for Korean fermentation companies ($1M+ annually per licensee) • Reduce capital requirements for GSH production (no specialized lysis equipment needed) • Position Korea as cost-competitive supplier to Asian and global markets • Job creation: manufacturing, quality control, sales positions at SME partners **Scientific Impact:** • Demonstrate viability of temperature-triggered secretion as general bioprocessing platform • Advance understanding of stress-induced metabolite export mechanisms • Publish findings in high-impact biotechnology journals • Train graduate students and postdocs in industrial biotechnology • Establish UIUC-Korea collaboration model for other technologies **Commercial Impact:** • Technology transfer to at least one Korean SME partner by project end • Licensing agreements generating royalty revenue for UIUC • Commercial-scale production capacity established within 2 years post-project • Market penetration in Korean cosmetics industry (K-beauty applications for GSH in skin products) • Export opportunities to Japan, China, Southeast Asia leveraging Korean manufacturing base **Sustainability Impact:** • More than 50% reduction in waste generation (no biomass disposal from lysis) • Lower energy consumption (elimination of cell disruption and extensive purification) • Reduced chemical usage in downstream processing • Model for "green" bioprocessing applicable to other products **Strategic Impact:** • Strengthen Korea-US research partnership and technology commercialization pipeline • Position Korean biotechnology sector in high-value biochemical production • Create template for university-industry collaboration in biotechnology • Demonstrate government R&D investment ROI through rapid commercialization pathway

6. References

(1) D. S. Laman Trip & **H. Youk**. Yeasts collectively extend the limits of habitable temperatures by secreting glutathione. *Nature Microbiology* (2020).

(2) D. S. Laman Trip, T. Maire, **H. Youk**. Slowest possible replicative life at frigid temperatures for yeast. *Nature Communications* (2022).